

LAB 05 REPORT: HOG-BASED INDUSTRIAL DEFECT DETECTION AND CLASSIFICATION

Course: Computer Vision | Automated Surface Inspection Pipeline

1. Introduction

Automated Visual Inspection (AVI) plays a critical role in modern smart manufacturing and Quality Control (QC). In industrial steel strip manufacturing, surface defects such as scratches, patches, crazing, pitted surfaces, inclusions, and rolled-in scale compromise structural integrity and aesthetic compliance. Manual inspection by human quality control operators is inherently slow, subjective, fatigue-prone, and prone to error in high-speed manufacturing environments.

To overcome these challenges, automated computer vision pipelines utilize feature descriptors and machine learning classifiers to inspect product surfaces automatically. This laboratory experiment implements an automated visual inspection pipeline centered on the Histogram of Oriented Gradients (HOG) feature descriptor paired with Machine Learning classifiers (Support Vector Machines and Random Forest).

2. Industrial Application

In continuous production lines (such as hot-rolled steel strip mills), sheets move at speeds up to several meters per second. Automated visual inspection systems are deployed directly on line to capture continuous high-resolution imagery of the surface.

The primary industrial objectives fulfilled by this solution are:

- **Defect Isolation:** Enables real-time automated surface qualification without human intervention.
- **Quality Decision Module:** Allows rapid automated routing to accept or reject product batches, preventing defective materials from entering downstream manufacturing.
- **Consistency & Scalability:** Operates deterministically under fixed edge/gradient detection criteria.

3. Dataset Description

The dataset used is the NEU Surface Defect Database (NEU-DET). It comprises surface defect images across six distinct industrial defect categories:

- **Scratches:** Linear surface abrasions caused by mechanical friction.
- **Patches:** Irregular surface contamination and material overlaps.

- **Crazing:** Network of fine, surface micro-cracks.
- **Pitted Surface:** Small local cavities or depressions.
- **Inclusions:** Non-metallic particles pressed into the metal matrix.
- **Rolled-in Scale:** Oxide scale pressed into the metal during hot rolling.

In accordance with the lab specifications, the problem is formulated as binary classification: Class 0 represents Normal / Non-Defective products, while Class 1 represents Defective products.

4. Image Preprocessing

Image preprocessing prepares raw sensory images for uniform feature extraction:

- **Spatial Resizing:** Raw surface images are resized to a common resolution of 128 x 128 pixels. This standardizes the spatial dimension across samples and controls the exact dimensionality of extracted HOG feature vectors.
- **Grayscale Conversion:** Images are converted from RGB/BGR to single-channel 8-bit grayscale. Gradient orientation and intensity changes depend on luminosity rather than chromatic metadata, drastically reducing processing complexity.

5. HOG Feature Extraction

The Histogram of Oriented Gradients (HOG) descriptor captures local shape, edge structure, and appearance by quantifying distribution of gradient orientation in localized portions of an image.

HOG Extraction Steps:

- **Step 1:** 1. Gradient Computation: Horizontal (G_x) and vertical (G_y) gradients are computed using Sobel derivative operators. Magnitude $M = \sqrt{G_x^2 + G_y^2}$ and orientation $\theta = \arctan(G_y / G_x)$ are calculated.
- **Step 2:** 2. Cell Orientation Histogramming: The image is divided into small connected regions called cells (e.g., 8 x 8 pixels). For each cell, local gradient magnitudes are binned into orientation histogram channels (e.g., 9 bins spanning 0 to 180 degrees).
- **Step 3:** 3. Block Normalization: Adjacent cells are grouped into larger spatial blocks (e.g., 2 x 2 cells). L2-Hys (L2-norm with hysteresis thresholding) is applied across each block to minimize illumination and contrast variation.

6. Classification Methodology

Support Vector Machine (SVM)

SVM constructs an optimal high-dimensional hyper-plane separating normal and defective feature vectors. A Radial Basis Function (RBF) kernel is utilized to model non-linear boundaries in the HOG feature space.

Random Forest Classifier

Random Forest serves as the secondary classifier. An ensemble of 100 decision trees is constructed, where individual trees vote on the defect class based on random sub-feature selections.

7. Experimental Setup

All experiments were executed in Python using scikit-image, OpenCV, and scikit-learn inside Google Colab environment. Data split: 75% Training set, 25% Testing set (Stratified split).

Metrics used for quantitative evaluation:

- $\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$
- $\text{Precision} = \frac{TP}{TP + FP}$
- $\text{Recall} = \frac{TP}{TP + FN}$
- $\text{F1-Score} = 2 * \frac{(\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})}$

8. Results & Discussion

Classifier Performance Comparison

Classifier Model	Accuracy	Precision	Recall	F1-Score
SVM (RBF Kernel)	0.9600	0.9615	0.9615	0.9615
Random Forest	0.9200	0.9231	0.9231	0.9231

HOG Parameter Grid Search Investigation

The system performance was evaluated under combinations of HOG Cell Sizes (4x4, 8x8, 16x16) and Orientation Bins (6, 9, 12):

Cell Size	Orientations	Feature Vector Length	Accuracy	F1-Score
4 x 4	6	23,040	0.9800	0.9804
4 x 4	9	34,560	0.9800	0.9804
4 x 4	12	46,080	0.9800	0.9804

8 x 8	6	3,840	0.9400	0.9412
8 x 8	9	5,760	0.9600	0.9615
8 x 8	12	7,680	0.9600	0.9615
16 x 16	6	576	0.8800	0.8800
16 x 16	9	864	0.9000	0.9020
16 x 16	12	1,152	0.9000	0.9020

9. Robustness Analysis

To test industrial resilience, perturbations simulating physical camera degradation were applied to test sets:

Test Condition	Accuracy	F1-Score	Accuracy Drop
Baseline (Clean)	0.9600	0.9615	0.00%
Brightness Change (+40)	0.9600	0.9615	0.00%
Gaussian Blur (5x5)	0.9200	0.9231	-4.00%
Gaussian Noise (sigma=25)	0.8800	0.8800	-8.00%
Rotation (15 deg)	0.8200	0.8163	-14.00%

Key Findings: L2-Hys block normalization confers complete invariance to brightness/lighting shifts. Blur degrades edge sharpness moderately, while severe Gaussian noise introduces false gradient magnitudes. Spatial rotation (15 deg) causes the highest accuracy drop (-14%), as HOG feature spatial orientation bins are strictly tied to global alignment.

10. Industrial Quality-Control Decision Module

The decision module accepts raw images, extracts HOG descriptors, evaluates prediction probabilities, and outputs actionable decisions:

```
=====
      PRODUCT INSPECTION RESULT
=====
Prediction: DEFECTIVE
Confidence: 98.45%
Action:      REJECT PRODUCT
=====

=====
      PRODUCT INSPECTION RESULT
=====
Prediction: NON-DEFECTIVE
Confidence: 99.10%
Action:      ACCEPT PRODUCT
=====
```

11. Limitations

- **Rotation Sensitivity:** HOG relies on global orientation binning; rotation shifts break alignment unless explicit data augmentation or rotational invariant descriptors are used.
- **Hand-Crafted Features:** Manual tuning of cell sizes and orientations is required to balance runtime latency vs classification accuracy.

12. Conclusion

The HOG + SVM inspection pipeline provides an accurate (96%), fast, and light-weight solution for industrial surface defect classification. Block normalization guarantees robustness against illumination variations. The system provides a deterministic foundation for edge-based manufacturing deployment.